FRESCO

A Fruit Freshness and Shelf-Life Estimation Platform

An Intelligent Shelf-Life Food Estimation System.

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The Problem: Food Waste

Why It Happens

❌ People lack reliable information about food condition

❌ Most fresh fruits have no expiration date at all

❌ Sensory inspection is subjective and unreliable

What Makes Freshness Assessment Difficult?

- **Problem 1: Subjective Assessment**

❌ Consumers rely on sight, touch, and smell

❌ Two people looking at the same fruit reach different conclusions

❌ Visible mould means contamination started 1-2 days earlier.

- **Problem 2: Ignorance of Storage Interactions**

❌ Bananas produce ethylene gas that spoils nearby fruits faster

❌ Storing bananas with strawberries reduces shelf life by 2-3 days

❌ Consumers are unaware of these invisible interactions

- **Problem 3: Inaccessible Solutions**

❌ Accurate systems require expensive sensors and cameras

❌ Hyperspectral imaging costs tens of thousands of dollars

❌ IoT sensor systems are impractical for households

A Smartphone-Based Freshness Assessment System

What FRESCO Does

✅ Identifies fruit type from a photograph

✅ Assesses visual freshness on a 0–1 scale

✅ Estimates remaining shelf life in days

✅ Warns about storage conflicts (ethylene)

✅ Recommends best storage method

✅ Tracks personal fruit inventory

- Key Feature: Accessibility
- No extra hardware required
- Just a smartphone camera

Aim & ObjectivesAim & Objectives

Aim

To design, develop, and implement a mobile-first web application that estimates fruit freshness and shelf life using only a smartphone camera.

.	Objective
1	Develop fruit identification using MobileNetV2 transfer learning.
2	Implement continuous freshness scoring (0.0–1.0).
3	Build shelf-life estimation engine using curated fruit data.
4	Deliver as Progressive Web : Application with inventory and ethylene checking.

Methodology

- **Phase 1: Literature Review**
- Review computer vision, transfer learning, post-harvest biology
- Identify datasets: Fruits-360, Kaggle Fresh/Stale

- **Phase 2: Architecture Design**
- Three-tier: Frontend (Next.js) → Backend (FastAPI) → Database (PostgreSQL)
- Seven-step scan pipeline

Phase 3: Implementation

> Train MobileNetV2 on Google Colab

> Build full-stack application

Phase 4: Testing

> Evaluate accuracy on held-out test set

> Validate shelf-life estimates

System Architecture

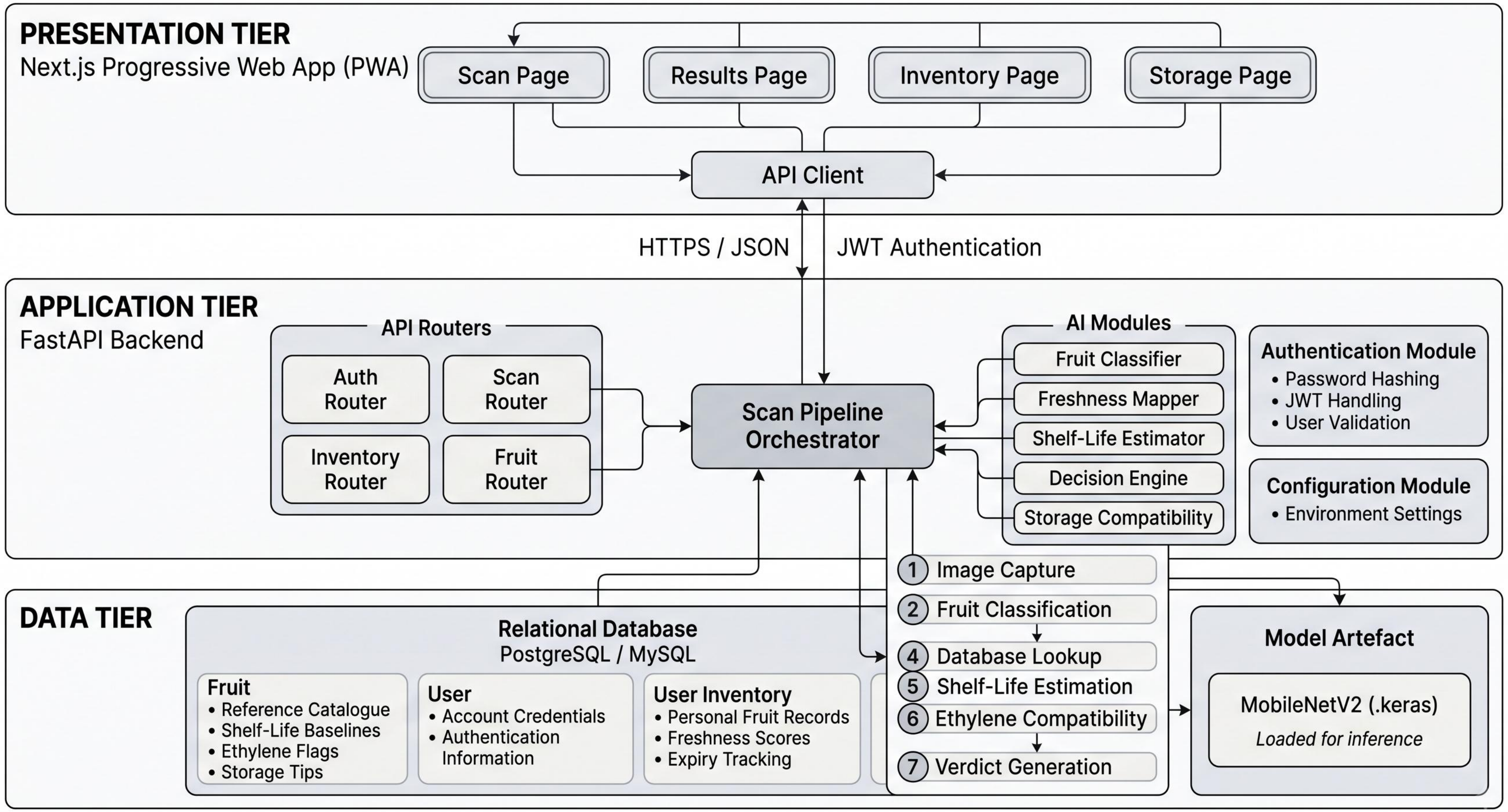

Figure 3.1: High-Level Architecture of the Fresco System

Five Freshness Categories

Score Range	Category	Action
0.80 – 1.00	Fresh	Normal storage
0.60 – 0.79	Slightly Aged	Eat soon
0.40 – 0.59	Aging	Use within 1-2 days
0.20 – 0.39	Old	Consume immediately
0.00 – 0.19	Spoiled	Discard

Results & Conclusion

Classifier Performance

✅ 89.3% top-1 accuracy on held-out test set

✅ 97.5% top-3 accuracy

✅ 200–500ms inference latency (after warm-up)

Key Strengths

✅ Hardware Accessibility: Works on any smartphone

✅ Ethylene Awareness: First consumer-facing implementation

✅ Interpretable: Clear recommendations with explanations

✅ Installable: No app store required (PWA)

Limitations

❌ Classifier covers 4 fruit types automatically (Apple, Banana, Carrot, Tomato)

❌ 38 fruits supported via manual selection

❌ Lighting sensitivity (known CNN limitation)

❌ No real-time environmental sensing